

THE ZERO PARADOX

ZP-A: Lattice Algebra

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This document is self-contained within abstract algebra. No topology, probability, or Hilbert space is imported. Every claim proved here is provable using only the tools of semilattice theory; the status of OQ-A1 points to other documents. Cross-framework connections are deferred to ZP-E.

Illustrated Companion: A paired ZP-A Illustrated Companion document provides concrete examples and visual intuitions for the results in this document. Examples are kept separate from the formal layers to distinguish illustrative material from proofs. The companion is a reading aid; no proof-critical judgements should be drawn from examples alone.

Metatheoretic Declaration — ZF + AFA

This document is stated over ZF + AFA (Zermelo–Fraenkel set theory with Aczel’s Anti-Foundation Axiom). AFA replaces the classical Axiom of Foundation and permits self-containing sets — in particular, sets satisfying $x = \{x\}$.

Scope: This declaration affects only CC-2 (Section V), which asserts $\perp = \{\perp\}$. All algebraic results in Sections I–IV are independent of AFA and hold in standard ZF.

Standard concrete models (power sets ordered by inclusion, real intervals ordered by max, etc.) satisfy A1–A4 but do not satisfy $\perp = \{\perp\}$. This is expected — they are models of the algebraic structure, not instantiations of the ZF + AFA metatheory. The self-containment of $\perp$ is a set-theoretic claim about what $\perp$ is, not an algebraic one.

The Axiom of Choice is not assumed. AFA is argued to be forced rather than chosen — see Section V and ZP-E Remark R-AFA for the argument.

I. Primitives and Axioms

1.1 Signature

The algebraic signature of the Zero Paradox state space is a triple: $(L, \vee, \perp)$

L is a non-empty set (the carrier set of states). $\vee : L \times L \rightarrow L$ is a binary operation called join. $\perp \in L$ is a distinguished constant called the bottom element.

Axiom Block A — Join-Semilattice with Bottom

A1 — Associativity: $(x \vee y) \vee z = x \vee (y \vee z)$ for all $x, y, z \in L$

A2 — Commutativity: $x \vee y = y \vee x$ for all $x, y \in L$

A3 — Idempotency: $x \vee x = x$ for all $x \in L$

A4 — Identity (Additive): $\perp \vee x = x$ for all $x \in L$

A4 is the load-bearing axiom. It makes $\perp$ the additive identity of the algebra: the element that contributes nothing to a join and is therefore present in every state as the neutral constituent.

II. The Induced Partial Order

2.1 Definition of $\leq$

Definition D1 — Lattice Order

For $x, y \in L$, define the relation $\leq$ by:

$$x \leq y \iff x \vee y = y$$

Proposition T1 — $\leq$ is a Partial Order

Reflexivity: $x \leq x$ — by A3, $x \vee x = x$. ✓

Antisymmetry: if $x \leq y$ and $y \leq x$, then $x \vee y = y$ and $y \vee x = x$. By A2, $y = x \vee y = y \vee x = x$. ✓

Transitivity: if $x \leq y$ and $y \leq z$, then $x \vee z = x \vee (y \vee z) = (x \vee y) \vee z = y \vee z = z$, so $x \leq z$. ✓

2.2 $\perp$ is the Least Element

Lemma T2 — $\perp$ is a Global Minimum under $\leq$

For all $x \in L$: $\perp \leq x$

Proof: By A4, $\perp \vee x = x$. By D1, this is the definition of $\perp \leq x$. ✓

T2 is the algebraic statement of the foundational claim: $\perp$ is not a void that states depart from — it is the minimum element that every state sits above. Since $\perp \leq x$ for all x , and join accumulates from the bottom, $\perp$ is algebraically present in every element of L .

III. The Additive Ontology

3.1 No Subtraction Operator

Remark R1 — Join-Semilattice vs. Lattice

A full lattice $(L, \vee, \wedge, \perp, \top)$ includes a meet operator $\wedge$ and a top element $\top$. The Zero Paradox restricts to the join-semilattice with bottom. The meet operator is excluded because it would allow state reduction — the removal of informational content from a state. The additive ontology requires that no operation decreases informational content.

3.2 Join is the Only State Transition

Definition D2 — State Transition

A state transition is any function $f: L \rightarrow L$ such that $x \leq f(x)$ for all $x \in L$.

Definition D2 — State Transition

Equivalently, for each $x \in L$, $f(x) = x \vee \alpha$ for some $\alpha \in L$.

Proof of equivalence:

($\Rightarrow$) If $x \leq f(x)$, then $x \vee f(x) = f(x)$ by D1. Take $\alpha = f(x)$: then $f(x) = x \vee \alpha$. ✓

($\Leftarrow$) If $f(x) = x \vee \alpha$ for some $\alpha \in L$, then $x \vee f(x) = x \vee (x \vee \alpha) = (x \vee x) \vee \alpha = x \vee \alpha = f(x)$ by A1, A3. By D1, $x \leq f(x)$. ✓

IV. Monotonicity of State Sequences

4.1 State Sequences

Definition D3 — State Sequence

A state sequence is a function $S: \mathbb{N} \rightarrow L$, written $(S_0, S_1, S_2, \dots)$, such that:

$$S_{n+1} = S_n \vee \alpha_n \text{ for some } \alpha_n \in L, \text{ for all } n \in \mathbb{N}$$

Remark R2 — Terminology: State Sequence and Ascending Chain

In the order-theory literature, a sequence (S_n) satisfying $S_n \leq S_{n+1}$ for all n is called an ascending chain. The term "state sequence" is used here in place of "ascending chain" to align with the state-transition framing of ZP-D and ZP-E, where the same structure is introduced as sequences of system states. The two terms denote the same mathematical object. Readers familiar with order theory should read "state sequence" as "ascending chain". For concrete illustrations, see the ZP-A Illustrated Companion.

Theorem T3 — State Sequences are Monotone

For any state sequence (S_n) satisfying D3: $S_n \leq S_{n+1}$ for all $n \in \mathbb{N}$

Proof: By D3, $S_{n+1} = S_n \vee \alpha_n$. By D1, $S_n \leq S_n \vee \alpha_n \iff S_n \vee (S_n \vee \alpha_n) = S_n \vee \alpha_n$. By A1, $(S_n \vee S_n) \vee \alpha_n = S_n \vee \alpha_n$. By A3, $S_n \vee S_n = S_n$. Therefore $S_n \vee \alpha_n = S_{n+1}$. ✓

Monotonicity is a theorem, not a postulate. It is derived from A1–A3 via D3.

4.2 The Initial State

Every state sequence begins somewhere. T2 establishes $\perp \leq S_0$ for any initialisation — the bottom element is always below the starting point, whatever that starting point is. But T2 does not fix where S_0 sits; a sequence could legitimately begin above $\perp$. CC-1 closes this gap: we commit to initialising at the minimum. This is a modelling choice at ZP-A scope. ZP-J restates it rather than removing it: in any AFAStructure lattice a sequence starts at a Quine atom exactly when it starts at $\perp$ (cc1_derived and t_exec_iff), so the choice of starting point is expressed through the Quine-atom role. It is still not forced: on any lattice with a point above $\perp$ (as AX-B1 supplies) a valid sequence can begin there, as the OntologicalStates example in ZeroParadox/Settheory/OntBridge.lean shows; on a one-point lattice every sequence starts at $\perp$.

Conditional Claim CC-1 — $S_0 = \perp$ | restated as an equivalence in ZP-J, not forced

We commit to initialising every state sequence at the minimum of L : $S_0 = \perp$. This is not derived from A1–A4 — it is a modelling choice.

Under CC-1 and T3: $S_0 = \perp \leq S_1 \leq S_2 \leq \dots$

Note: By T2, $\perp \leq S_0$ for any initialisation — this holds unconditionally from A4. CC-1 strengthens this to equality: $S_0 = \perp$. The commitment is not needed to establish $\perp \leq S_0$; it is needed to fix the starting point precisely.

Status: CONDITIONAL CLAIM — a modelling commitment at ZP-A scope. ZP-J restates it: given `AFAstructure`, `cc1_derived` proves that a state sequence starting at a Quine atom starts at $\perp$, and `t_exec_iff` gives the converse, so the two starting conditions are one. On a lattice with a point above $\perp$ a state sequence can begin there (T2 fixes only $\perp \leq S_0$; witness in `ZeroParadox/Settheory/OntBridge.lean`), so the equivalence does not remove the choice of starting point; it expresses it through the Quine-atom role.

V. The Self-Containment of $\perp$

5.1 Foundational Characterisation

The axioms A1–A4 establish $\perp$ as the additive identity and algebraic minimum of L . The following forced metatheoretic commitment characterises its set-theoretic nature. R3 provides a structural route to DA-1 in ZP-E: CC-2 establishes that $\perp$ has no external interpreter position, which — conditional on D7’s exhaustive static/executing dichotomy (ZP-E) as background — eliminates the static-description state for $\perp$. See R3 for the full dependency note.

Foundation note: The framework is stated over $ZF + AFA$ (Zermelo–Fraenkel set theory with Aczel’s Anti-Foundation Axiom). The classical Axiom of Foundation is replaced by AFA, which permits self-containing sets. This replacement is not an arbitrary modelling choice: $\perp = \{\perp\}$ is a member of itself ($\perp \in \perp$), which the Axiom of Foundation (Regularity) forbids — under a well-founded membership relation no set is self-membered (`no_quine_atom`, `ZeroParadox/Settheory/Wall.lean`) — so a Foundation universe cannot host $\perp$, and AFA is the anti-foundation setting that permits it. R3 and ZP-C L-INF corroborate the non-well-founded character of $\perp$ rather than independently prove the exclusion. See ZP-E Remark R-AFA for the full cross-framework argument. The Axiom of Choice is not assumed.

Forced Metatheoretic Commitment CC-2 — Self-Containment of $\perp$

The null state $\perp$ is its own extension: the collection of all objects bearing the structural property of $\perp$ is $\perp$ itself.

Formally: $\perp = \{\perp\}$

Under $ZF + AFA$, $\perp$ is a Quine atom — a set satisfying $x = \{x\}$. By set extensionality, any infinite collection of objects all indistinguishable under the structural property of $\perp$ collapses to $\perp$ itself. There is no multiplicity, only $\perp$.

Forced Metatheoretic Commitment CC-2 — Self-Containment of $\perp$

This is a Forced Metatheoretic Commitment, not a freely chosen modeling decision. Foundation cannot host $\perp = \{\perp\}$: the bottom is a member of itself, and Regularity forbids self-membership (`no_quine_atom`). The Foundation-freeness of any host of $\perp$ is forced and machine-checked (`QuineHost`; see ZP-E Remark R-AFA); what remains a commitment rather than a theorem is only that a Quine atom and its uniqueness are the right requirements to demand — this is not derived from A1–A4 at the algebraic level. The named falsifier is correspondingly narrow: a differently-motivated requirement set that hosts $\perp$ without demanding a unique Quine atom would reopen that choice; it would not touch the forcing, which is settled.

Status: FORCED METATHEORETIC COMMITMENT — Foundation-freeness of any host of $\perp$ is forced and machine-checked (`QuineHost`); only the requirements-choice (a unique Quine atom) remains a commitment, not a derivation. Algebraic fixed-point content formally verified in ZFC by ZP-J (`ScaleBridge`); the literal set-membership $\perp \in \perp$ is outside Lean scope (metatheoretic).

Cross-framework note: The replacement of Foundation by AFA is not an arbitrary choice — $\perp = \{\perp\}$ is a member of itself ($\perp \in \perp$), which Regularity forbids (`no_quine_atom`), so ZF + Foundation cannot host it. Foundation and AFA are dual framings of the same object: Foundation excludes the Quine atom; AFA uniquely permits it. AFA is the forced metatheoretic replacement; the specific form $\perp = \{\perp\}$ is the minimal Quine atom consistent with A4. R3 and ZP-C L-INF corroborate $\perp$'s non-well-founded character; they are not independent proofs of the exclusion. See ZP-E Remark R-AFA for the full cross-framework argument.

Lean 4 scope — three distinct layers. (1) Set-theoretic: the claim $\perp = \{\perp\}$ as a set cannot be realized in Lean's type theory (CIC is well-founded by construction). This remains a prose-level commitment in ZF + AFA. (2) AFAStructure context (the typeclass taken as a hypothesis): ZP-J defines `IsQuineAtom` as the lattice-theoretic analog — `IsQuineAtom q := selfMem q  $\wedge$   $\forall x$ , selfMem x  $\rightarrow$  x = q`. Within an `AFAStructure` instance, `t_exec_iff` proves `IsQuineAtom q  $\leftrightarrow$  q =  $\perp$` (`SetTheoryAFA.lean`); ZP-K's `da1_closed_concrete` closes this concretely on `MachinePhase`. These results take the `AFAStructure` typeclass as a hypothesis, which every ZP-A lattice can supply trivially, so the hypothesis carries no AFA set-theoretic content of its own. (3) ZFC-clean (no AFA import): `ScaleBridge.lean` proves the fixed-point content in standard ZFC — `selfMem_eq_singleton_free` and `z2_selfMem_singleton` establish $\{x : \mathbb{Z}_2 \mid 2x = x\} = \{0\}$. This result is loop-free from AFA. The set-theoretic interpretation connecting layer 3 to layer 1 remains outside Lean scope.

Remark R3 — CC-2 Eliminates the Static-Description State for $\perp$

A self-containing object has no external interpreter by structure: $\perp = \{\perp\}$ is its own interpretation. A description requires a describer distinct from the thing described; CC-2 admits no such distinction for $\perp$. Under the Turing model framework (D7, ZP-E), which partitions machine configurations into static-description states and executing states, the absence of any external interpreter position means $\perp$ cannot occupy D7's static-description category.

Dependency note: The inference from "no external interpreter" to "necessarily executing" uses D7's exhaustiveness in the final step — that static-description and executing are the only two categories. D7 (ZP-E) supplies this exhaustiveness as the shared background framework. R3 provides the structural argument for why $\perp$ engages D7's transition; it does not independently derive DA-1 without D7. All three DA-1 paths in ZP-E share D7 as background. Their independence is among their arguments — CC-2/R3 (structural), L-INF (informational), K-incompressibility (AIT) — not from D7 itself.

VI. OQ-A1 — Sufficiency of Monotonicity

OQ-A1 — Sufficiency of Monotonicity [OQ-A1b CLOSED by A1-A4; OQ-A1a: no reason to restrict to join-irreducibles (well-founded carriers); placement in the ordinal tower, for a monotone map sending the tower's stages to c_0 , conditional on the alignment hypothesis $h\varepsilon_0$ (ε_0 sent to c_1)]

Is the monotonicity constraint (T3) sufficient to characterise all valid state sequences, or are additional axioms required?

OQ-A1a: Is there algebraic reason to restrict α_n to join-irreducible elements (not expressible as joins of strictly smaller elements)?

OQ-A1b: Does the open-ended semilattice (without top element T) permit unbounded ascending chains?

Status: OQ-A1b CLOSED by A1-A4: yes. The natural numbers under max are a model, and every n has a first distinct state $n + 1$ there, so first steps do not give the ascending chain condition. (On the two-state carrier every chain is bounded; that is a fact about that carrier.)

OQ-A1a: no algebraic reason to restrict α_n to join-irreducible elements. On a chain the restriction is vacuous (every element above $\perp$ is join-irreducible); on a branching carrier it is not (the top of the subsets of a two-element set is the join of two smaller states); and in a well-founded carrier every increment is a finite join of join-irreducibles (Birkhoff 1937; see Chakir-Pouzet, arXiv:0812.2300, Thm 2.1(c); exists_supIrred_decomposition), so restricting changes the number of steps, not the set of reachable states. Beyond well-founded carriers this is not measured.

Separately, where the step is placed in the ordinal tower is conditional on the alignment hypothesis: for a monotone map sending the tower's stages to c_0 , given $h\varepsilon_0$ (ε_0 sent to c_1), nothing below ε_0 fires, because ε_0 is the supremum of the stages, and every fixed point of $\alpha \mapsto \omega^\alpha$ fires, because ε_0 is the least of them (snap_unconditional, hfp_from_epsilon_zero, epsilon0_min_eq_max). $h\varepsilon_0$ ($\varphi \varepsilon_0 = c_1$; in the $\mathbb{Z}_2$ chart, snapEmbed ($\varphi \varepsilon_0$) = 0) is the snap's occurrence at ε_0 , taken as a hypothesis: of the infinitely many admissible firing points that monotonicity and tower alignment (every tower stage sent to c_0) leave open, $h\varepsilon_0$ selects the least, and so fixes φ uniquely (ZeroParadox/Ordinal/Incompleteness.lean § II). Whether Classical.choice is forced by the metric collapse is a separate open question (ZeroParadox/Ordinal/SyntacticCollapse.lean).

No step is guaranteed. A state sequence moves only upward (T3) and never returns (R1, t_snap_irreversible). Where the next increment is already absorbed ($S_n \vee \alpha_n = S_n$), the state stays the same, which T-SNAP permits (tsnap_holds_but_nothing_moves); different runs need not reach the same height. If from some step x on every new increment is already absorbed, then S_x is the supremum of the run: the value that run reaches in the limit. Short of a top, reaching it cannot be confirmed at any finite step; a run that reaches a top state (c_1 on the two-state carrier) is known there to stay.

Reading: ε_0 is both at once for the ordinal tower: the least fixed point of $\alpha \mapsto \omega^\alpha$ and the supremum of its stages (epsilon0_min_eq_max).

OQ-A1 — Sufficiency of Monotonicity [OQ-A1b CLOSED by A1-A4; OQ-A1a: no reason to restrict to join-irreducibles (well-founded carriers); placement in the ordinal tower, for a monotone map sending the tower's stages to c_0 , conditional on the alignment hypothesis $h\epsilon_0$ (ϵ_0 sent to c_1)]

Note on minimal element above $\perp$: Bounded chains do not in themselves guarantee a minimal element above $\perp$. In a dense structure, for any $\epsilon > 0$, $\epsilon/2$ also exists — chains may be bounded yet have no first step. That is not a hypothetical: it is ZP-F, where the halving argument is proved for every field carrying a compatible linear order (f_snap_impossible, ZeroParadox/Reals/OrderedField.lean), and is stated there as the direct negation of AX-B1 (axb1_fails_in_ordered_field). The closure of OQ-A1b does not bound ascending chains, and a bound would not establish that ϵ_0 exists as a specific minimal non-null element either. In the lattice chart that existence rests on the same binary existence commitment (AX-B1); in the ordinal chart ϵ_0 is proved to be both the least fixed point of $\alpha \mapsto \omega^\alpha$ and the supremum of the tower's stages (epsilon0_min_eq_max), and a monotone map that sends those stages to c_0 and ϵ_0 to c_1 fires nowhere below ϵ_0 , by the supremum face, and at every fixed point, by the least-fixed-point face (snap_unconditional, hfp_from_epsilon_zero). $h\epsilon_0$ ($\varphi \epsilon_0 = c_1$; in the $\mathbb{Z}_2$ chart, snapEmbed ($\varphi \epsilon_0$) = 0) is the snap's occurrence at ϵ_0 , taken as a hypothesis: of the infinitely many admissible firing points that monotonicity and tower alignment (every tower stage sent to c_0) leave open, $h\epsilon_0$ selects the least, and so fixes φ uniquely (ZeroParadox/Ordinal/Incompleteness.lean § II). Whether Classical.choice is forced by the metric collapse is a separate open question (ZeroParadox/Ordinal/SyntacticCollapse.lean). ZP-B's 2-adic structure places 0 at infinite valuation and every non-zero element at finite valuation; what it proves there is that the gap at 0 is clopen (T3) and the return across it irreversible (C3). The non-zero elements still come arbitrarily close to 0 ($\|2^n\|_2 = 2^{-n}$), so there is no closest non-zero element to serve as a first step. This is outside the scope of A1–A4.

VII. Boundary Conditions

Export	Status / Receiving Document
$(L, \vee, \perp)$ as join-semilattice	Derived (A1–A4) — ZP-D: algebraic structure of state space
$\leq$ partial order (D1, T1)	Derived — ZP-D: ordering on states
Monotonicity of state sequences (T3)	Derived from A1–A3 — ZP-D: state layer ordering
$\perp$ as global minimum (T2, CC-1)	Derived (T2) / Conditional (CC-1) — ZP-E: ontological grounding claim. CC-1 is restated in ZP-J as an equivalence (cc1_derived, t_exec_iff), not forced.
$\perp = \{\perp\}$ self-containment (CC-2, R3)	Forced Metatheoretic / Remark — ZP-E: structural route to eliminating static-description state for $\perp$, given D7 exhaustiveness as background. Fixed-point content verified in ZFC by ZP-J.
No subtraction / additive ontology (R1)	Structural — ZP-C: no operation may reduce informational content
OQ-A1 — increment selection	OQ-A1b CLOSED by A1–A4; OQ-A1a: no reason to restrict to join-irreducibles (well-founded carriers); placement in the ordinal tower, for a monotone map sending the tower's stages to c_0 , conditional on the alignment hypothesis $h\epsilon_0$ (ϵ_0 sent to c_1)

VIII. Validation Status

Component	Status / Notes
A1–A4 join-semilattice axioms	Valid — Axioms; self-contained
$\leq$ partial order (D1, T1)	Valid — Derived from A1–A3
$\perp$ as least element (T2)	Valid — Derived from A4 and D1
Additive ontology / no subtraction (R1)	Valid — Structural; signature restriction
State transition as join (D2)	Valid — Defined; consistent with signature
Monotonicity of state sequences (T3)	Valid — Derived from A1–A3 and D3
CC-1: $S_0 = \perp$	CONDITIONAL CLAIM — modelling commitment at ZP-A scope. ZP-J restates it as an equivalence (a Quine-atom start is a $\perp$ start: <code>cc1_derived</code> , <code>t_exec_iff</code>); the starting-point choice is not forced.
CC-2: $\perp = \{\perp\}$	Forced Metatheoretic Commitment — AFA over Foundation argued to be structurally required; fixed-point content verified in ZFC by ScaleBridge
ZF + AFA foundation (no AC)	Meta-theoretic — framework-wide; required for CC-2
OQ-A1: Sufficiency of monotonicity	OQ-A1b CLOSED by A1–A4; OQ-A1a: no reason to restrict to join-irreducibles (well-founded carriers); placement in the ordinal tower, for a monotone map sending the tower's stages to c_0 , conditional on the alignment hypothesis $h\varepsilon_0$ (ε_0 sent to c_1)